

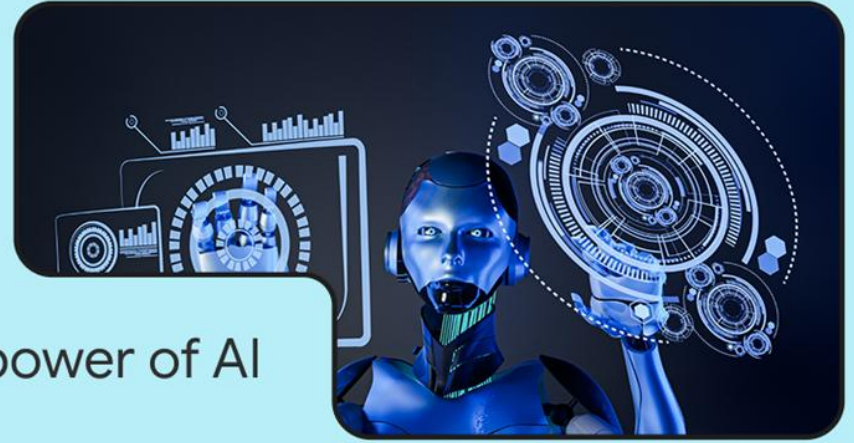


Google Developer Group
On Campus

TechSprint



Leveraging the power of AI



Team Details :

- a. Team name: TEAM COSMOS
- b. Team leader name: Om Shinde
- c. Problem Statement : Our problem statement is a user-centric access problem caused by language and digital literacy barriers in government service delivery.

Brief about your solution and problem statement addressing

★ PROBLEM STATEMENT

A significant portion of citizens, especially campus support staff and nearby local communities, remain **excluded from government welfare schemes** due to **information accessibility barriers**. Although many schemes exist, their details are published as lengthy and complex policy documents in English or formal Marathi, making them difficult to interpret for digitally low-literate users. Existing digital platforms primarily act as information repositories rather than guidance systems, resulting in low awareness, incorrect applications, and underutilization of public welfare programs.

★ Brief about the Solution

Sarathi is an AI-driven, voice-first assistance platform that bridges the gap between government policies and citizens. The solution leverages **OCR, Large Language Models, and conversational AI** to extract user demographics from identity documents, analyze eligibility against official government scheme documents, and deliver **accurate, personalized guidance** in Marathi or Hindi. By transforming complex policy PDFs into simple, actionable audio instructions, Sarathi ensures **inclusive access, reduced dependency on intermediaries, and improved scheme adoption** among underprivileged communities.

Opportunities

a.How different is it from any of the other existing ideas?

- > Existing solutions only **list schemes or provide links**
- > They require users to **read complex PDFs and websites**
- > Sarthi uses a **voice-first AI approach**
- > Automatically **matches schemes based on user eligibility**
- > Explains benefits and next steps in **local languages**
- > Designed specifically for **digitally low-literate users**

Aspect	Existing Solutions	Sarthi
User Interaction	Text-based	Voice-based (Marathi/Hindi)
Information Format	Long PDFs & links	Simplified audio guidance
Eligibility Check	Manual	AI-driven automatic matching
User Literacy Needed	High	Very low
Personalization	Same info for all users	User-specific schemes
Role	Information portal	Intelligent guidance system

b.How will it able to solve the problem?

Sarthi addresses the problem by **eliminating language, literacy, and complexity barriers** that prevent citizens from accessing government welfare schemes. Instead of expecting users to read and interpret policy documents, the system **automates understanding and decision-making** using AI technologies.

The solution works by extracting user demographic information from identity documents using OCR and collecting minimal additional inputs through a voice-based interaction. This data is then matched against official government scheme documents processed by a Large Language Model, ensuring accurate eligibility determination. Finally, Sarthi delivers **simplified, personalized, and actionable guidance** in local languages through audio, enabling users to understand benefits and next steps without external assistance.

By converting complex policies into **human-centric, voice-driven guidance**, Sarthi ensures higher awareness, correct applications, and improved adoption of government schemes among underserved communities.

List of features offered by the solution?



List of Features Offered by the Solution

•Voice-First User Interface

Enables hands-free interaction in Marathi and Hindi, eliminating the need for reading or typing.

•ID-Based Automatic Data Extraction

Uses OCR to extract demographic details from Aadhaar or Ration Card images.

•AI-Driven Eligibility Detection

Matches user data with government scheme eligibility criteria using Large Language Models.

•Simplified Policy Interpretation

Converts complex government policy documents into concise, easy-to-understand explanations.

•Personalized Scheme Recommendation

Suggests the most relevant government scheme for each user based on their profile.

•Audio-Based Guidance

Delivers scheme benefits and application steps through clear voice instructions.

•Minimal User Input Requirement

Requires only one ID upload and a few simple voice responses.

•Scalable and Language-Agnostic Design

Can be extended to additional regional languages and schemes in the future.

Google Technologies used in the solution?

Cloud Vision OCR: For reading local ID cards.

Gemini : reads" ID photos, simplifies complex policy PDFs into local dialects, and matches users to the right schemes through voice-based conversation.

Dialogflow: To manage the conversational flow in Marathi/Hindi.

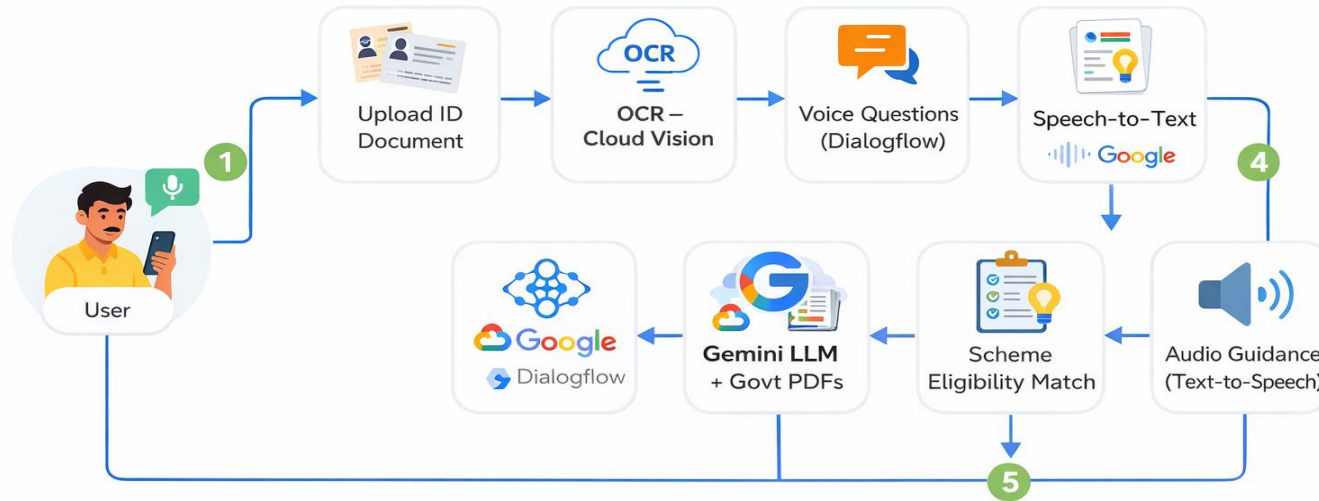


Google Cloud Vision



Process flow diagram or Use-case diagram

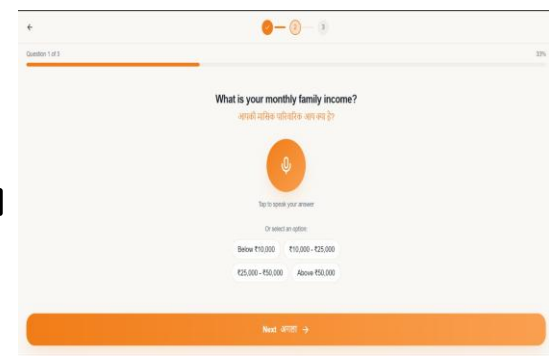
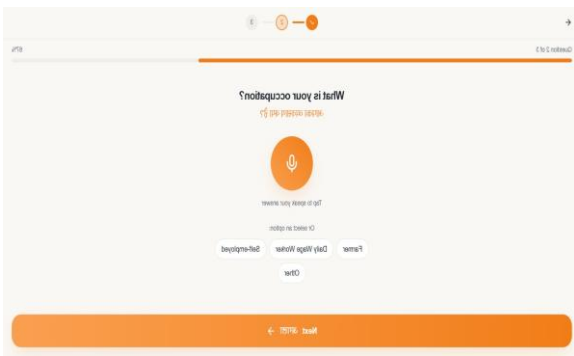
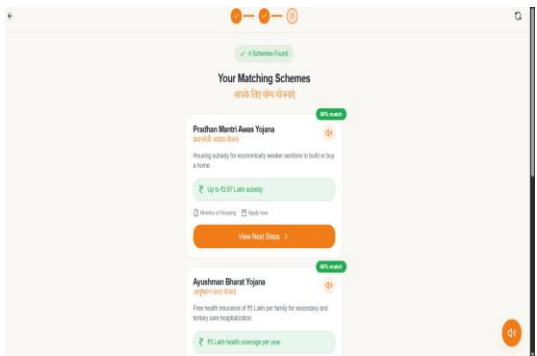
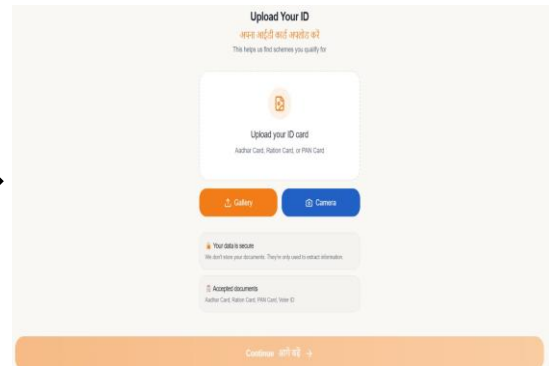
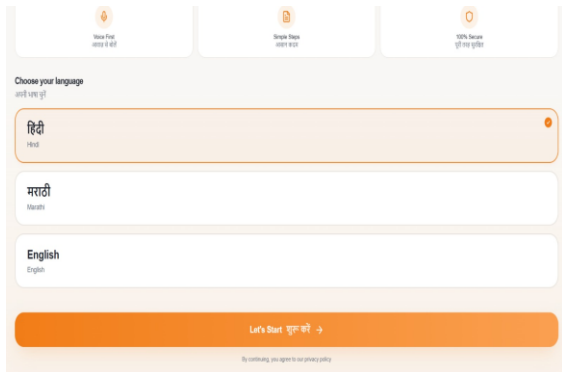
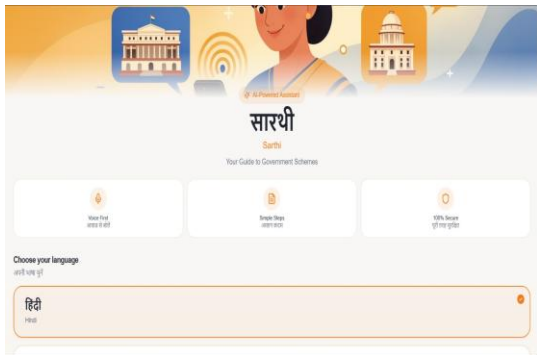
Process Flow of the Proposed Solution (Sarathi)



- User opens the app → uploads an ID → answers simple voice questions → gets eligible scheme details via audio output



Snapshots of the MVP



Additional Details/Future Development (if any)

Why this is innovative :

It breaks the **"Complexity Barrier."** Most apps just list links; this app acts as a specialized legal consultant for the underprivileged.

 **Real-world impact** Directly connects users to financial aid (e.g., MAJYAV); potentially secures **thousands of Rupees** in benefits per user.

RAG: The AI in our prototype does not guess scheme details. It first retrieves information from official government PDFs and then generates responses, ensuring correctness and trust.

 **Hackathon MVP scope** A voice-enabled interface that takes one ID photo and returns one matched scheme with the "Next Steps" explained in audio.

 **Future potential** Can be adopted by NGOs or the local Municipal Corporation (PMC) to digitize social welfare outreach.



Conclusion

Sarthi demonstrates how Artificial Intelligence can be used to create inclusive, accessible, and impact-driven digital solutions. By combining OCR, conversational AI, and Large **Language Models**, the platform transforms complex government policies into **simple, voice-based guidance** for citizens who need it the most.

The solution not only improves **awareness and accessibility** of **government welfare schemes** but also reduces dependency on intermediaries and ensures **accurate scheme adoption**. With scalability across languages and regions, Sarthi has strong potential for real-world deployment through NGOs, municipal bodies, and government **partnerships**.

Sarthi bridges the gap between policy and people — ensuring no eligible citizen is left behind.



THANK-YOU

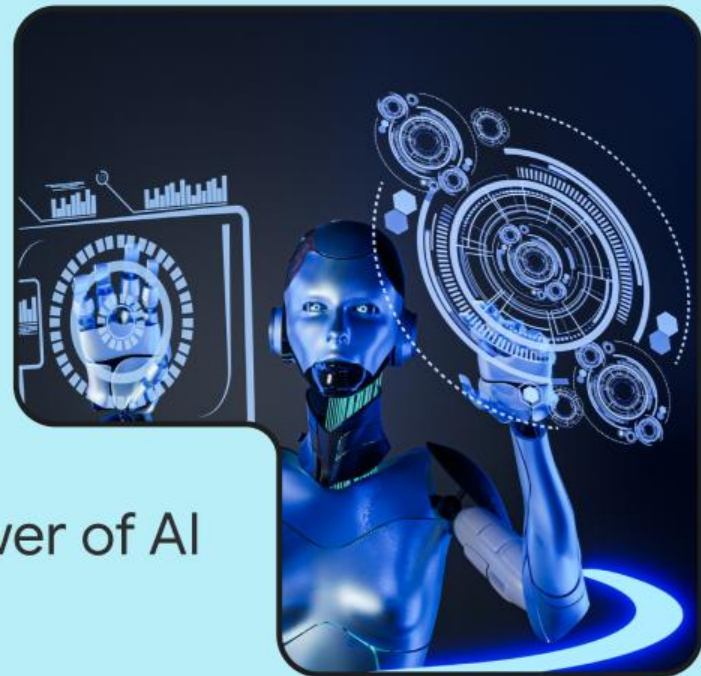


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Thank you!

